

Machine Learning Classification of Variable Stars

ADRIEN BONANSEA,^{1,*} DUSAN ISKIC,^{1,*} AND AMITAI F. M. HEYL^{1,2,*}

1

²University of Toronto

ABSTRACT

The automatic classification of periodic variable stars is of immense value for many reasons, including finding standard candles, determining stellar properties accurately, and studying stellar life cycles. Additionally, missions like TESS and Gaia, which are unified in the MMU dataset, have provided millions of data points to study them. However, with this wealth of information comes the need to classify the different variable star types quickly. This project implements a machine learning approach to address this challenge. We developed a TESS processing pipeline that utilizes Lomb-Scargle periodograms to automatically extract periodicity features and filter for variability. These features are then used to manually classify the stars and feed the data to the programs training set. Our results demonstrate a functional feature extraction pipeline and a Random Forest classifier capable of distinguishing variable types, providing a scalable framework for analyzing the MMU dataset. Initial testing with a small data set of about 150 yielded a 63% accuracy. Expanding the training set to 688 and implementing rigorous data curation yielded a significantly improved accuracy of 76.81%.

1. INTRODUCTION

Variable stars can provide critical insights into various stellar properties—such as mass, radius, luminosity, temperature, structure, material composition, and overall evolution (Percy, 2004). Periodic variable stars, such as Cepheids and RR Lyrae, are especially important as standard candles for measuring cosmic distances and as probes of internal stellar structure. This report takes advantage of all-sky surveys like TESS (Transiting Exoplanet Survey Satellite) and Gaia, using time-domain datasets containing millions of light curves across the sky. With time-series data of variable stars, we can analyze the light curve and determine properties such as amplitude and period using a Fourier transform, similar to Szklenár et al. (2022).

We use TESS light curves to find the periods with *Lomb-Scargle*, a built-in Astropy function that helps detect periods in uneven observations. By analyzing each light curve with our Python function, we determine the amplitude and period for each star.

Next, we developed a variable star classification system based on methodologies from Szklenár et al. (2022) and Hoffman et al. (2009). To achieve this, we integrated the $G_{BP} - G_{RP}$ color index from Gaia with time-domain features from TESS—specifically period, amplitude, and light curve morphology. Additionally, we utilized the Simbad database to cross-reference and validate specific targets against known classifications.

Using this feature set, we generated a labelled training dataset to train a Random Forest Classifier, enabling the automated categorization of variable stars at scale. This is immensely valuable due to the enormous amount of data provided by GAIA and TESS that would be unrealistic and inefficient to fully manually classify.

1.1. Star Types

CLASSICAL CEPHEID: CLASSICAL CEPHEIDS AND TYPE-II CEPHEIDS

These stars are known as yellow supergiants. “Their temperatures are 6000–8000K, and their periods are 1–100 days or more” (Percy, 2007, ch. 6). ‘Classical’ Cepheid stars are a subcategory of Cepheids which are young stars more massive than the Sun, and conversely, Type-II Cepheids are old stars less massive than the Sun (Percy, 2007, ch. 6).

Cepheid brightness variation is known for its periodicity. The period and amplitude range is “periods of 1 to 70 days with an amplitude variation of 0.1 to 2.0 magnitudes” (Harvard-Smithsonian Center for Astrophysics, 2015).

* y

“The simplest form of pulsation is radial pulsation – a simple, spherically symmetric in-and-out expansion and contraction... Large amplitude pulsating stars – Cepheids, RR Lyrae stars, and Mira stars – pulsate primarily in radial modes” (Percy, 2007, ch. 6).

δ SCUTI STARS, HIGH-AMPLITUDE δ SCUTI VARIABLE STARS

These are pulsating stars of type A and F. They have short periods and small amplitudes (Percy, 2007, ch. 6). Dwarf Cepheids are categorised as high-amplitude Delta Scuti stars (Percy, 2007, ch. 6). Small amplitudes mean they are usually discovered through means of photoelectric or CCD photometry (Percy, 2007, ch. 6). Delta Scuti stars have a typical period of 0.02 to 0.3 days (Harvard-Smithsonian Center for Astrophysics, 2015).

RR LYRAE AB TYPE: FUNDAMENTAL-MODE, FIRST-OVERTONE AND DOUBLE-MODE

The RR Lyrae star’s variability was observed around the turn of the 20th century by Edward Pickering after its discovery by Wilhelmina Fleming (Newton, 2012, ch. 6). They show short periods of about 0.567 day and high amplitude pulsations (Newton, 2012; Percy, 2007, ch. 6).

They exhibit the following features: “0.5 solar mass and have a short pulsation period of 0.05 to 1.2 days and amplitude variations of 0.3 to 2 magnitudes” (Harvard-Smithsonian Center for Astrophysics, 2015).

They are defined by their evolutionary status; having exhausted all of their hydrogen fuel, they have now switched to burning helium in the core (Percy, 2007, ch. 6). The temperature of stars like this varies, but luminosity is consistent, and they lie on the “horizontal bar” of the H-R diagram (Percy, 2007). “Their narrow range of temperatures and luminosities explains their small range of periods” (Percy, 2007, ch. 6).

ECLIPSING BINARIES

A lot of stars are part of a binary or even multiple star system, but the large majority of these are wide binary systems. In these binary systems, the two stars are actually “hundreds of times greater than the average distance of the Earth from the Sun” (Percy, 2007, ch. 5). If they are close to the Earth, we can see them as visual binaries. It is very hard to observe any variation in brightness for geometric reasons (Percy, 2007, ch. 5). Close binary stars, defined by Percy (2007), can vary in brightness because their geometry allows for it.

The dips in the light curve are created when one star passes in front of Earth’s field of view of the other star. The light curve has two dips in it: the deeper one is called the primary, and the shallower one is the secondary eclipse (Percy, 2007, ch. 5). Thus, an interval will include two eclipse dips, the secondary and primary (Percy, 2007, ch. 5). Percy (2007) shows an example of this with the light curve of Algol (β Persei). The period of close eclipsing stars is “generally in the range of 0.25 to ten days” (Percy, 2007, ch. 5).

ERUPTIVE VARIABLE STARS: CATAclysmic AND FLARE STARS

“Eruptive variables are considered to include any variables which flare up relatively quickly and fade more slowly” (Percy, 2007, ch. 7). This can include flare stars (Percy, 2007, ch. 7). Flare stars will randomly and unpredictably shoot up in brightness, specifically in the UV band, and then gradually fall (Percy, 2007, ch. 7). These types of stars are usually dwarf K and M stars; our Sun could be considered a flare star (Percy, 2007, ch. 7).

When Percy looks at a typical flare, the rise time is a fraction of a minute, and the decay is a few minutes, but for some stars, this may be hours, and “the total energy released is 3×10^{28} to 3×10^{34} ergs; the luminosity of the star itself is 10^{28} to 10^{33} ergs/sec” (Percy, 2007, ch. 7).

Cataclysmic variables (CV) are another eruptive variable that can take many forms but all follow the same principle: two close binary stars where one is a main-sequence star fills the area around it with matter and then transfers it to its partner star, a compact object, which then fills its accretion disc with the matter (Percy, 2007, ch. 7).

γ CASSIOPEIA VARIABLE STARS

Gamma Cassiopeiae variables, known in Percy (2007) as a type of ‘Be type’ star, are defined as non-supergiant stars that have temperatures between 10,000K and 30,000K and emission lines in their spectra. These stars are very luminous, and many of them can be seen with the naked eye (Percy, 2007, ch. 9). ‘Be’ star brightness varies on many different timescales, from decades to fractions of a day.

Percy goes on to say that different time variations are the result of different effects: between weeks and decades can be a result of dispersal of the disc; variations between months and years are due to wave motions in the disc; days to weeks are associated with binary motion and interaction; timescales smaller than a day can be due to non-radial pulsation and/or perhaps rotation.

Table 1. Characteristics of different types of variable stars.

Type of Variable Star	H–R Position	Amplitude	Period
Cepheid (Classical + Type-II)	Instability strip (yellow supergiants)	0.1–2.0 mag	1–70 d (some to 100+ d)
Delta Scuti; high-amplitude Delta Scuti	Near main sequence (A–F type)	Small	0.02–0.3 d
RR Lyrae ab: fundamental/1st-overtone/double-mode	Horizontal branch	0.3–2 mag	0.05–1.2 d (typ. 0.567 d)
Eclipsing binaries	—	Brightness dips during eclipses	0.25–10 d
Eruptive (cataclysmic and flare stars)	K–M dwarfs; white-dwarf binaries	Rapid, unpredictable	Rise: < 1 min; decay: min–hours
Gamma Cassiopeiae variables	Upper main sequence (B-type)	Varies (disc/pulsation)	Fractions of a day to decades
Rotational variables	Any (depends on type)	Depends on brightness regions	Stellar rotation period
Mira (long-period) stars	Asymptotic giant branch (red giants)	≥ 2.5 mag	80–1000 d

ROTATIONAL VARIABLE STARS

When a star is rotating in the Earth’s field of view, it can have brighter and less bright areas that get shown during its rotation; this is what then leads to a varying brightness over time (Percy, 2007, ch. 4). The period of brightness variation is the period of rotation. The variation can be seen both photometrically and spectroscopically (Percy, 2007, ch. 4).

MIRA, LONG-PERIOD STARS

These stars are known by their longer periods of 80 to 1000 days and deviate in brightness by more than 2.5 magnitudes (Harvard-Smithsonian Center for Astrophysics, 2015). Mira is a stage for most mid-size main sequence stars as they evolve to red giants (Harvard-Smithsonian Center for Astrophysics, 2015).

EARLY CHARACTERIZATION OF STAR TYPES

Based on our research gathered up until the midterm, we were able to compile a table (Table 1).

2. DATA

2.1. TESS

The data used in our light curve modeling comes from TESS from the Multimodal Universe dataset. TESS, or Transiting Exoplanet Survey Satellite, is a space telescope that is part of NASA’s exploratory mission, launched in 2018. It was made to search for transiting exoplanets in orbit around over 200,000 stars. Because of this, it has a huge amount of time-series data for many stars’ magnitudes, which is exactly what we need to conduct our light curve modeling.

2.2. Gaia

To complement the photometric data, we integrated astrometric and color information from the European Space Agency’s Gaia mission (launched 2013). Specifically, we extracted the $G_{BP} - G_{RP}$ color index to constrain stellar temperature and evolutionary state. As of 2025, Gaia has mapped the motion, luminosity, and composition of over two billion objects (European Space Agency, n.d.) providing the necessary color-magnitude context to distinguish between variable star classes on the Hertzsprung-Russell diagram.

3. METHODS

3.1. Lomb–Scargle

Our first tool we apply is the Lomb-Scargle Periodogram through astropy. This allows us to find periods from data with inconsistent spacing. This was required for our TESS data as it does have gaps in its data that a standard Fourier transform cannot handle. However, the Lomb-Scargle method only identifies the best-fitting period. This creates limitations for light curves that exhibit two oscillation wavelengths. To compensate for this, we introduced a variable, `is_binary`, to help both the machine-learning model and the manual classification identify cases where two or more periodic components are present.

3.2. Phase Folding

Once we’ve gotten the period with Lomb-Scargle, we phase fold using that period to both improve the signal to noise ratio, as well as see if the period is a good fit. As if the period is a poor match the phase-folded graph will appear random and not have a notable pattern. This allowed us to test if the Lomb-Scargle method worked and as we can observe in Figure 1 the folded light curve clearly showed a consistent period.

3.3. Star Filtering

Now using these tools as well as some helpers we made, we developed a pipeline to determine which stars from the data are variable, as many are not. To do this we first used our `clean_para` and `rem_outliers` helper functions to clean the data by removing outliers and normalizing the light curve. Then we run our `compute_features` function which uses Lomb-Scargle to get the period, as well as amplitude and the signal-to-noise ratio. We use these to sift through a large set of stars to see which are good potential targets. This pipeline identified two likely variable stars, which are shown in Figures 1 and 2. These plots show the raw light curve, its Periodogram, and the phase-folded curve, confirming they are indeed variable.

We found that by setting a minimum Lomb-Scargle power of 0.3, a minimum amplitude of 0.001, and a minimum signal-to-noise of 3, we were able to see most of the high variability. These 3 variables let us look at the shape and the power of the signal compared to the noise, to ensure the oscillations were not created by noise.

After loading 10,000 stars, we realized that multiple entries shared the same star ID. After checking several of these cases, we found that some corresponded to the same star but had different light curves due to the data being taken at different times. However, this was not true for all repeated stars, as some had exactly the same light curves and data as the original entry. Due to this complication, we decided to keep only the initial pass for each star ID. This ensures that we do not classify the same star multiple times and bias the machine-learning classifier.

3.4. Gaia Matching

As the stars were formatted as a list of dictionaries, we were able to examine all filtered stars with high variability and use their RA and DEC values to determine their positions. Instead of performing HEALPix matching, we chose to use the `astroquery` library because of its efficiency and its ability to retrieve data directly from online astronomical catalogs. Using `astroquery`, we queried the Gaia DR3 database with each star's RA and DEC coordinates and obtained the relevant information. This included the Gaia source ID, parallax, parallax error, `bp-rp` color, and several additional features that were less important for our analysis.

In the cases where the `bp-rp` colors are missing from the Gaia catalog, we skip the star and remove it from the high variability stars.

3.5. Classification method

We created a manual classification system which can be broken down into a couple of steps. The first step or resource we consulted was the SIMBAD archive of stars, here we could input the Gaia star ID and see what type of variable star it was, this usually only worked for long period or eclipsing binary stars.

If the SIMBAD archive search failed we would then resort to a manual classification based first upon the light curve shapes which we would compare to examples of variable star light curves from the OGLE Atlas of Variable Star Light Curves. This narrows down the classification but does not provide a definite result. After the light curve we go into classification based on parameters such as colour, amplitude and period.

Using the *OGLE Atlas of Variable Star Light Curves* and the textbook *Percy, J. R. (2007). Understanding Variable Stars. Cambridge University Press*, we were able to refine our previous classification system, producing a larger and more functional system that applies to a broader range of stars.

Variable Type	Amplitude	Period	Color (BP–RP)
Classical Cepheids (Type I)	0.1–2.0	1–70 d	0.5–1.2 (yellow)
Type II Cepheids	0.3–1.2	1–50 d	0.3–1.0
BL Herculis Stars	0.3–1.0	1–4 d	0.2–0.6
Delta Scuti Stars	0.003–0.5	0.02–0.3 d	0.0–0.5 (white–yellow)
RR Lyrae Stars	0.3–2	0.05–1.2 d	0.1–0.6
Long Period Variables (LPVs)	Varies	30–1000 d	1.5–4.0 (deep red giants)
Mira Variables	≥ 2.5	10–1000 d	≥ 2.0 (very red)
Small Amplitude Red Giant	0.003–0.1	10–100 d	1.0–3.5
Eclipsing Binaries	Brightness dips	0.25–10 d	Any (depends on stars)
Gamma Cassiopeiae Variables	Various	0.05–3650 d	–0.2 to 0.2 (blue)
Rotational Variables	Depends on star-spots	Stellar rotation	Depends on spectral type
BY Draconis Variables	0.03–0.30	0.2–20 d	1.0–3.0
Red Giant Rotational Variable	0.005–0.05	10–200 d	1.2–3.0 (red)
Long-Period Variable (Mira, SR, OSARG)	0.02–2.5	20–1000 d	1.2–4.0

Table 2. Characteristics of Various Types of Variable Stars

For the manual classification, we are assuming that the interpretation made by our group members of the light curve and data is accurate. We also treat SIMBAD classifications as definitive. It is likely that we misclassified some star types when handling such a large catalogue, especially in borderline cases where a certain star met the qualifications for more than one type due to both human bias, error and possible errors in SIMBAD. Manual classification is very time consuming and worked for us because we used a smaller training set, but if we wished to scale up, we would need to find a more efficient classification system.

3.6. Choosing Machine learning method

The Random Forest Classifier is well-suited to our goals because of how it organizes and interprets data. It begins by taking our sample dataset and creating thresholds for each parameter. Then, it builds multiple decision trees based on these parameters and thresholds, passing each star through every tree. Finally, it averages the resulting classifications to determine the most accurate class for each object.

This approach is particularly effective for our data since not all variable stars will perfectly fit within every threshold. The Random Forest Classifier can still correctly classify a star even if one parameter deviates from the expected range.

4. RESULTS

The TESS pipeline was successfully implemented allowing the processing, cleaning, and variability filtering for each star. This pipeline was able to both determine which stars are variable and then plot them to verify their variability. Their high Lomb-Scargle (LS) power is a strong indicator of this variability.

4.1. Light Curve Plots

A good example of why we used both the light curve and the parameters is the classification difference between star 229911482 (Figure 1) and star 229948027 (Figure 2). Their parameters are quite similar (Table 2) yet star 229911482 was classified as a Small Amplitude Red Giant (OSARG), and star 229948027 as a Red Giant Rotational Variable (RGV). This distinction would not have been possible if we hadn't examined each star's light curve to see that their shapes are different.

Property	229911482	229948027
TESS Light Curve Properties		
Period (days)	10.688709	10.369746
Amplitude	0.013288	0.009816
Gaia DR3 Properties		
BP-RP color	1.544	2.064

Table 3. Comparison of TESS and Gaia DR3 properties for stars 229911482 and 229948027.

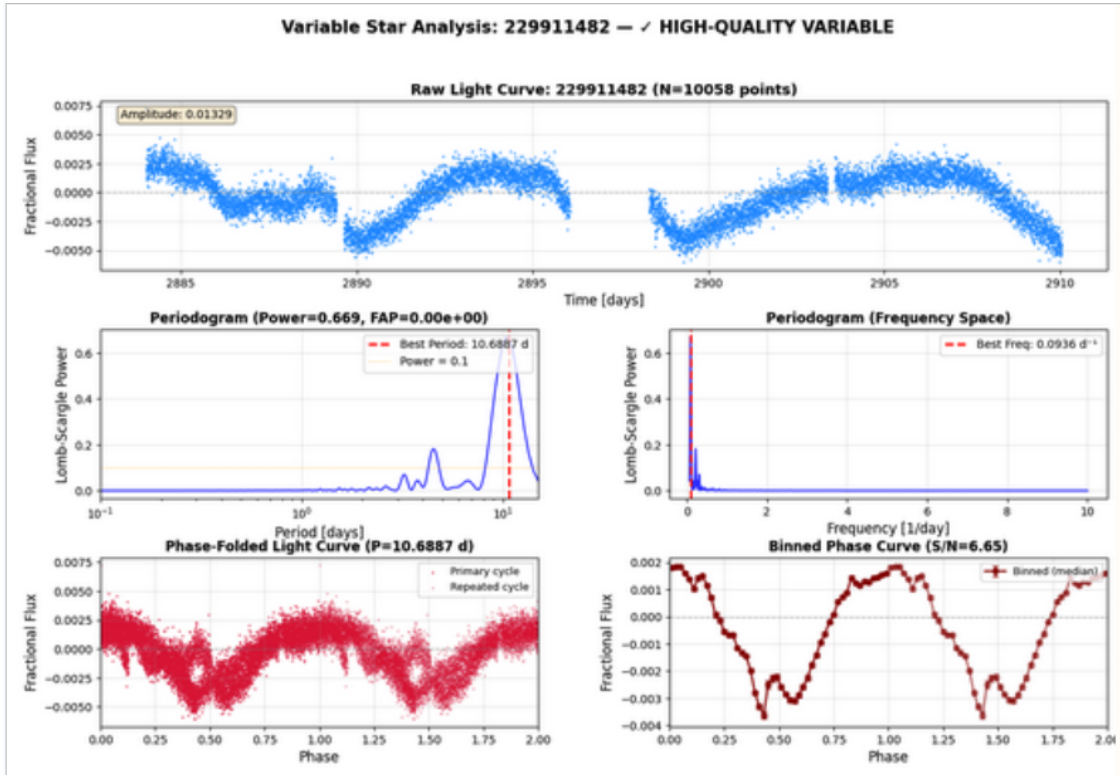


Figure 1. Diagnostic plot for variable star 229911482. Showing both a clear period in both the raw and folded data, and an obvious Periodogram spike. Additionally the LS power was 0. 537

 HIGH-QUALITY VARIABLE #1

Variable Star Analysis: 229948027 — ✓ HIGH-QUALITY VARIABLE

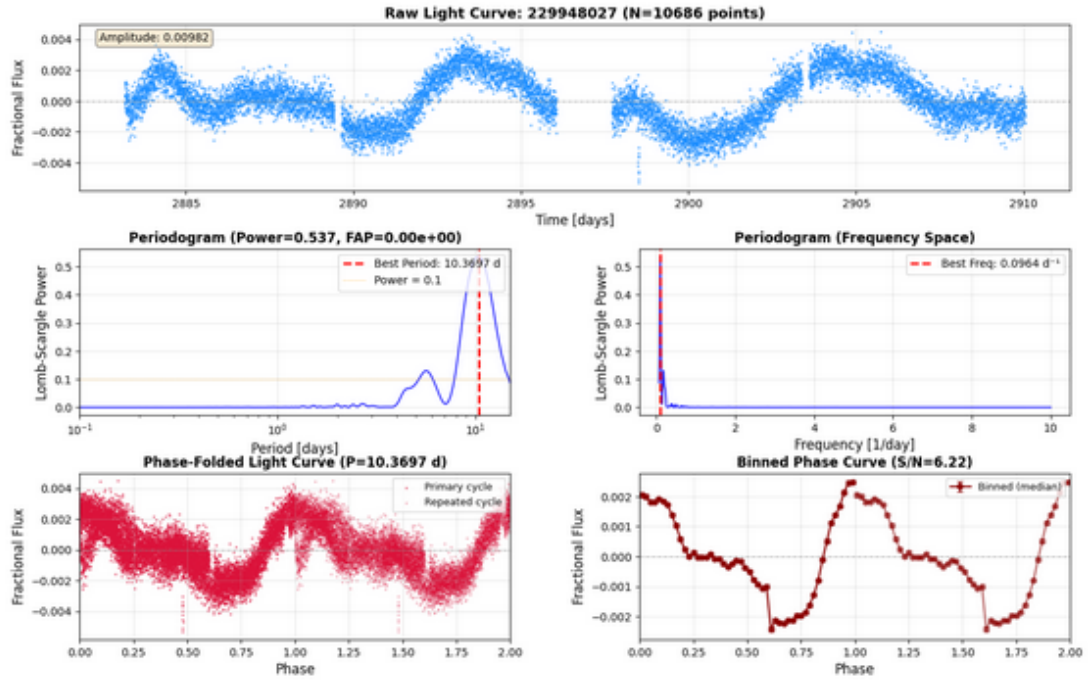


Figure 2. Diagnostic plot for variable star 229948027.

4.2. Classification results

In this section, we present the classification results and the number of stars in each category:

Class	Count
RGV	219
ROT	160
EB	73
BY	72
OSARG	71
GDOR	26
DSS	19
DSCT	14
LPV	12
CC1	9
MV	8
NOID	5
SPB/ROT	3
EB/ROT	3
DSCT/EB	3
CC2	2
BL	2
BY/EB	2
RRL	1
SPB	1
EB/DSCT	1
BYS	1

Table 4. Class distribution of variable star types in the dataset. Total training set star count = 550

As we can see from the table above, we had a relatively uneven distribution of stars. To train the ML model, we had to drop star types that had 3 or fewer appearances. Class imbalance affects the accuracy of the model as it can skew results. If a star has 3 or fewer appearances, the model cannot learn well predict when this star type will appear.

4.3. Random Forest Algorithm Results

Although our dataset is relatively small, we wanted to evaluate how the machine learning model performs with the available data. Given the limited training set and strong class imbalance, an accuracy of 76.81% is consistent with expectations and demonstrates that the model captures broad morphological differences but struggles with underrepresented classes.

4.3.1. Example Machine Learning Output

In this section, we illustrate why the random forest algorithm was chosen. Table 5 shows the predicted class probabilities for an example star that is known to be a binary system.

Class	Probability	Percentage
EB	0.9550	95.5%
ROT	0.0250	2.5%
GDOR	0.0100	1.0%
OSARG	0.0067	0.7%
LPV	0.0033	0.3%
BY	0.0000	0.0%
CC1	0.0000	0.0%
DSCT	0.0000	0.0%
DSS	0.0000	0.0%
MV	0.0000	0.0%
NOID	0.0000	0.0%
RGV	0.0000	0.0%

Table 5. Predicted class probabilities for the object. The model predicts the object to be an eclipsing binary (EB) with high confidence.

From the table, we observe that the algorithm predicts the EB class with a probability of 95.5%, followed by ROT at 2.5% and GDOR at 1% and some other that have very small percentages. This indicates that the model is functioning as expected and is correctly identifying the most likely class for this star.

4.3.2. ML Output

This section presents the overall performance of the random forest algorithm, summarized in the classification report shown in Table 6.

Class	Precision	Recall	F1-score	Support
BY	0.80	0.57	0.67	14
CC1	0.00	0.00	0.00	2
DSCT	0.50	0.33	0.40	3
DSS	1.00	0.50	0.67	4
EB	0.94	1.00	0.97	15
GDOR	1.00	1.00	1.00	5
LPV	0.00	0.00	0.00	2
MV	0.00	0.00	0.00	2
NOID	1.00	1.00	1.00	1
OSARG	0.40	0.14	0.21	14
RGV	0.71	0.95	0.82	44
ROT	0.79	0.94	0.86	32
Accuracy		0.77		138
Macro Avg	0.59	0.54	0.55	138
Weighted Avg	0.72	0.77	0.72	138

Table 6. Classification report for the variable star classifier, showing precision, recall, F1-score, and support for each class.

The overall accuracy of the algorithm is 76.81%, which is above our initial target of 60%, but still relatively low. As shown in the previous section, the model itself functions correctly; however, several factors likely contribute to the limited performance.

One major issue is the imbalance and small size of the dataset. Certain classes are underrepresented or did not appear at all in the training data. This prevents the algorithm from learning their patterns effectively as is the likely reason as to our error rate being 23.19%

4.3.3. Feature Importance Report

Feature	Importance
bp_rp	0.3513
period	0.1413
is_binary	0.1273
amplitude	0.1059
kurtosis	0.0917
second_harmonic	0.0756
asymmetry_ratio	0.0537
rise_time_fraction	0.0532

Table 7. Feature importance ranking from the classification model.

We can see that the **bp_rp** color index is the most informative feature. This is expected, as many stars share similar variability characteristics but differ significantly in their color, making **bp_rp** a strong discriminator between classes. The **is_binary** feature also plays an important role, helping to distinguish binary systems from other types of variable stars, especially in cases where features such as **kurtosis** are less effective at identifying complex or multi-peaked light curves.

The features **rise_time_fraction** and **asymmetry_ratio** have relatively low importance. This likely occurs because they capture similar information to **second_harmonic** and **kurtosis**, which already describe the overall shape and structure of the light curves more effectively. This suggests that future versions of our algorithm may benefit from removing or reworking these lower-impact features, or from expanding the feature set to include more informative descriptors of variability.

5. DISCUSSION

Overall, our pipeline was successful in analyzing the photometric properties of stars in the TESS dataset and effectively filtering for variable star candidates. By automating the feature extraction, we were able to scale our analysis from a small sample to a collection of hundreds of variable star candidates. This expanded dataset enabled the preliminary training of our Random Forest Classifier, which establishes a baseline for automated classification with 76.81% accuracy. We anticipate the model’s accuracy will improve significantly as we continue to expand the training set to our target of 1,000+ objects.

5.1. Comparison to Papers

The first version of our machine learning model has a significantly lower accuracy than the one presented in the paper: 76.81% compared to their reported 95%. This difference can be explained in part by the size and balance of the training sets. The authors used a dataset of 2,000 to 3,000 stars, with a minimum of 50 stars per variability type, whereas our training set contains only 160 stars, with an additional 40 in the test set.

In the paper, the authors also cross-matched stars using the ASAS-SN, Gaia DR3, and TESS-EB catalogs, rather than relying on a manual classification system.

In Richards et al. (2011), they found that the random forest model is among the fastest and most accurate. When using the random forest method, they did find common classification errors such as: “Lambda Bootis and Beta Cephei are frequently classed as Delta Scuti; Wolf-Rayet, Herbig AE/BE, and S Doradus stars are usually classified as Periodic Variable Super Giants, and T Tauri are often misclassified as Semiregular Pulsating Variables” (Richards et al. 2011).

The Richards et al. (2011) paper also introduced a new system for classification using Hierarchical Classification. In this system, stars are broken up into groups similar to a phylogenetic tree. Each type can be traced back to a common ancestor at the top. As you work down, the larger group is split again and again into smaller and smaller groups that are more specific until you reach the nodes, which are the final classification of the star. Decisions made at the top of the tree have the largest room for error, and there is less and less uncertainty as you get closer to the nodes. This helps explain errors between closely related variable star types in our model.

In Szklenár et al. (2022), the authors developed a multi-input neural network that took advantage of image classification. They input an image of the star’s light curve to the Convolutional Neural Network and use that as a parameter

to classify the star. This is different from our methodology, in which we extract physical and numerical features—such as period, amplitude, and color—and use them as inputs for random forest classifiers. We did use the light curve image for manual classification, and in some instances it was needed to distinguish between two separate classes that shared similar physical properties.

5.2. *Next Steps*

Future work could include further expanding the training set, testing deep learning architectures such as Convolutional Neural Networks (CNNs) to analyze light curve morphology directly, similar to the approach used in Szklenár et al. (2022), and incorporating additional spectral features from Gaia to refine classification boundaries.

In the Richards et al. (2011) paper, the authors experimented with random forest classification and used inverse class-size weighting. While the overall error increased by 23%, the model performed much better for smaller classes. The number of correctly classified Lambda Bootis stars increased from 1 to 7, Beta Cephei stars from 5 to 23, eclipsing variables from 6 to 11, and Gamma Doradus stars from 8 to 15 (Richards et al. 2011).

Another approach would be to use preclassified star types to train our model, reducing our reliance on manual classification. This would likely remove a significant amount of uncertainty and result in a more standardized model.

5.3. *Paper impact*

Our result accuracy of 76.81% for our machine learning model makes common mistakes, and with its 0% precision rate for Cepheids, it cannot reliably identify these crucial objects used for distance estimation. Our model can be useful for large-scale population studies of common variable types where trends would matter more than the classification accuracy. Our high precision for eclipsing binaries suggests that our model may provide a particular utility.

6. CONCLUSION

In this study, we successfully developed a pipeline to extract and classify variable star data by integrating photometry from the TESS mission with astrometric parameters from Gaia DR3. We utilized the Lomb–Scargle periodogram to analyze light curves, identifying periodic signals and filtering for variability. By cross-referencing these targets with the SIMBAD archive, as well as utilizing manual classification, we constructed a labelled dataset to serve as ground truth for our classifier.

Once classified, we trained a random forest algorithm on approximately 550 variable stars and tested it on around 138 stars. Once classified, we trained a random forest algorithm on approximately 550 variable stars and tested it on around 138 stars. This resulted in an overall accuracy of 76.81%, which is below our desired final performance, likely due to many of the star types being underrepresented and having to be left out of the ML training. For the training set, the ML was able to sort through 138 stars in a couple seconds while the manual classifications had a rate of around 60 stars an hour. This is a drastic improvement and shows the power of using automation for this task as we can both automatically find variable stars, and then identify their type.

For further refinement, we could use preclassified star types, a CNN model instead, and experiment with class-size weighting.

REFERENCES

- Center for Astrophysics — Harvard & Smithsonian. (n.d.). Transiting Exoplanet Survey Satellite (TESS). <https://www.cfa.harvard.edu/facilities-technology/telescopes-instruments/transiting-exoplanet-survey-satellite-tess>
- European Space Agency. (n.d.). Gaia. https://www.esa.int/Science_Exploration/Space_Science/Gaia
- Gao, X., Chen, X., Wang, S., & Liu, J. (2024). Classification of periodic variable stars from The Transiting Exoplanet Survey Satellite (TESS). *The Astrophysical Journal Supplement Series*, 276(1), 57. <https://doi.org/10.3847/1538-4365/ad9dd6>
- Harvard-Smithsonian Center for Astrophysics. (2015, March 9). Pulsating variable stars and the Hertzsprung–Russell (H–R) diagram. https://chandra.harvard.edu/edu/formal/variable_stars/bg_info.html
- NASA. (2025, August 28). Diagram of an exoplanet phase curve. NASA Science. <https://science.nasa.gov/asset/webb/diagram-of-an-exoplanet-phase-curve/>
- Newton, E. (2012, July 4). RR Lyrae variables: History written in the stars. *Astrobites*. <https://astrobites.org/2012/07/04/rr-lyrae-variables-history-written-in-the-stars/>
- OGLE Atlas: Optical Gravitational Lensing Experiment. (n.d.). OGLE Atlas of Variable Star Light Curves. Retrieved December 2, 2025, from <https://ogle.astrouw.edu.pl/atlas/>
- Percy, J. R. (2004). Variable stars research resources home. <https://www.astro.utoronto.ca/~percy/>
- Percy, J. R. (2007). *Understanding variable stars*. Cambridge University Press.
- Richards, J. W., Starr, D. L., Butler, N. R., Bloom, J. S., Brewer, J. M., Crellin-Quick, A., Higgins, J., Kennedy, R., & Rischard, M. (2011). On machine-learned classification of variable stars with sparse and noisy time-series data. *The Astrophysical Journal*, 733(1), 10. <https://doi.org/10.1088/0004-637X/733/1/10>
- SIMBAD database: Wenger, M., Ochsenbein, F., Egret, D., Dubois, P., Bonnarel, F., Borde, S., Genova, F., Jasniewicz, G., Laloe, S., Lesteven, S., & Monier, R. (2000). The SIMBAD astronomical database: The CDS reference database for astronomical objects. *Astronomy & Astrophysics Supplement Series*, 143, 9–22. Retrieved December 2, 2025, from <https://simbad.u-strasbg.fr/simbad/sim-basicIdent=m33&submit=SIMBAD+search>
- Szklenár, T., Bódi, A., Tarczay-Nehéz, D., Vida, K., Mező, Gy., & Szabó, R. (2022). Variable star classification with a multiple-input neural network. *The Astrophysical Journal*, 938(1), 37. <https://doi.org/10.3847/1538-4357/ac8df3>
- Use of AI: AI was used to correct English:
<https://chatgpt.com/share/69140e33-22c0-8005-a22e-339f1485a836>
<https://chatgpt.com/share/692fbff5-5bd4-8009-b1dd-0aa4d7bf7c6d>